Bioinspiration & Biomimetics

# Development of a biologically inspired multi-modal wing model for aerial-aquatic robotic vehicles through empirical and numerical modelling of the common guillemot, *Uria aalge*

To cite this article: Richard J Lock *et al* 2010 *Bioinspir. Biomim.* **5** 046001

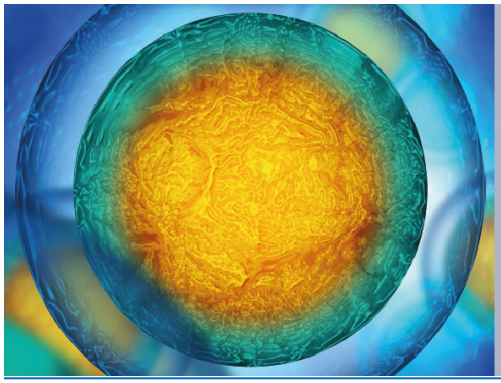
View the article online for updates and enhancements.

Related content

- Multi-modal locomotion: from animal to application
R J Lock, S C Burgess and R Vaidyanathan
- Gyroscopic sensing in the wings of the hawkmoth *Manduca sexta*: the role of sensor location and directional sensitivity
Brian T Hinson and Kristi A Morgansen
- Asymmetries in wing inertial and aerodynamic torques contribute to steering in flying insects
Mark Jankauski, T L Daniel and I Y Shen

Recent citations

- Hybrid parameter identification of a multi-modal underwater soft robot
F Giorgio-Serchi *et al*
- Perspectives on biologically inspired hybrid and multi-modal locomotion
K H Low *et al*
- Launching the AquaMAV: bioinspired design for aerial–aquatic robotic platforms
R Siddall and M Kova



Biophysical Society IOP ebooks™

Your publishing choice in all areas of biophysics research.

Start exploring the collection–download the first chapter of every title for free.

This content was downloaded from IP address 14.139.37.108 on 11/10/2021 at 15:00

Bioinsp. Biomim. **5** (2010) 046001 (15pp) doi:10.1088/1748-3182/5/4/046001

# Development of a biologically inspired multi-modal wing model for aerial-aquatic robotic vehicles through empirical and numerical modelling of the common guillemot, *Uria aalge*

**Richard J Lock**[1], **Ravi Vaidyanathan**[1,2], **Stuart C Burgess**[1] **and John Loveless**[3]

[1] Department of Mechanical Engineering, Bristol Robotics Laboratory, University of Bristol, Bristol BS8 1TR, UK
[2] Department of Systems Engineering, Naval Postgraduate School, Monterey, CA 93930, USA
[3] Department of Civil Engineering, University of Bristol, Bristol BS8 1TR, UK

E-mail: richard.lock@bristol.ac.uk, s.c.burgess@bristol.ac.uk, r.vaidyanathan@bristol.ac.uk, rvaidyan@nps.edu and J.Loveless@bristol.ac.uk

Received 16 June 2010
Accepted for publication 12 October 2010
Published 8 November 2010
Online at stacks.iop.org/BB/5/046001

**Abstract**
The common guillemot, *Uria aalge*, a member of the auk family of seabirds exhibits locomotive capabilities in both aerial and aquatic substrates. Simplistic forms of this ability have yet to be achieved by robotic vehicle designs and offer significant potential as inspiration for future concept designs. In this investigation, we initially investigate the power requirements of the guillemot associated with different modes of locomotion, empirically determining the saving associated with the retraction of the wing during aquatic operations. A numerical model of a morphing wing is then created to allow power requirements to be determined for different wing orientations, taking into account the complex kinematic and inertial dynamics associated with the motion. Validation of the numerical model is achieved by comparisons with the actual behaviour of the guillemot, which is done by considering specific mission tasks, where by the optimal solutions are found utilizing an evolutionary algorithm, which are found to be in close agreement with the biological case.

(Some figures in this article are in colour only in the electronic version)

## 1. Introduction

Advances in fabrication, sensors, electronics and power storage have made possible the development of a range of small robotic vehicles capable of aerial, terrestrial and even aquatic locomotion [1–3]. Additionally, insights into animal locomotion have significantly improved the mobility and stability of these vehicles [4]. At this time, however, the challenges of robotic design for vehicles capable of locomotion through multiple media (e.g. air, water, land) have received considerably less attention in the literature [5]. In particular, the design tradeoffs involved between transitioning from aerial to aquatic locomotion modalities for an autonomous robot have yet to be addressed by the research community. Potential applications of such a vehicle include the offshore oil industry, where the vehicle could fly from remote oil rigs, subsequently diving underwater to inspect pipe work and joints. Another application involves maritime counter-terrorism operations where a boarding crew could launch a small robot to conduct both aerial and aquatic inspection/surveillance of a larger

 © 2010 IOP Publishing Ltd Printed in the UK

cargo ship. These are just some of the potential tasks that could be completed; the development of aerial/aquatic multi-modal vehicles would therefore represent a generational leap in robotic utility.

Although no mature examples of small robots capable of operation in air and water exist today, many animals have developed multi-use mechanisms that are efficient in both terrestrial and aquatic substrates; studying the multi-modal nature of animals can clearly open potential avenues of research that will greatly improve locomotive performance of currently designed vehicles [6]. Optimization of competing design tradeoffs between locomotion mechanisms is a particularly ripe area for study.

As inspiration, and indeed motivation for this research, avian species have been identified as suitable candidates for study due to the scalability of the mechanisms they use compared with those exhibited by insects. By no means are the authors saying additional insights cannot be learnt from studying insects, simply that at this time focus will be on avian solutions.

Birds have achieved aerial/aquatic multi-modal locomotion via two distinct means: by using independent mechanisms for each or a single technique adapted for operations in both mediums. Foot-propelled divers tend to be slower bottom feeders and typically use their wings to aid with manoeuvrability [7, 8], although diving grebes have been shown to use a more efficient lift-based strategy with feet alone [9]. However, birds have achieved higher levels of performance in terms of speed, manoeuvrability and endurance by means of a flapping technique using the wings [10–12]. For this reason the authors have decided to focus their research on the use of a multi-modal wing rather than two distinct mechanisms. This will introduce the need to compromise the overall wing design, but it is felt that required levels of performance will be obtainable with this chosen route.

This work therefore documents research to further our understanding of basic avian mechanisms enabling both aerial and aquatic locomotion, with specific focus on wing behaviour to generate propulsive forces in both mediums. Empirically determined values associated with wing profile drag are found for static wing representations of the wing shape used by the common guillemot, *Uria aalge*, in air and water highlighting the potential saving in profile drag achieved by the folding mechanism. Furthermore a detailed numerical model of a simplified wing profile required to perform aerial and aquatic locomotion is developed to capture the varying kinematics experienced by the flapping wing during aquatic locomotion, whilst introducing inertial contributions unaccounted for in the initial empirical study. An evolutionary algorithm is implemented to balance competing design tradeoffs captured in the model and validate the results based on desired performance characteristics through comparisons with the natural (avian) system. Contrasting characteristics are then investigated for an extended and retracted wing orientation in order to investigate the implications of wing shape on overall power requirements of the aquatic flapping motion taking into account variations in kinematics and inertial and hydrodynamics. Folding the wing akin to the guillemot is found to consistently reduce the overall power requirements associated with the aquatic flapping mechanism.



**Figure 1.** Guillemot during aerial and aquatic locomotion (adapted from an unpublished BBC footage).

## 2. Avian inspired aerial/aquatic vehicle

It is our vision to perform basic research that elucidates understandings of how an entity may fly great distances, and also dive under the water and by simply morphing the shape of its wing enable the use of the same propulsion mechanism in a fluid of completely different density. This is, in essence, what the auk family of seabirds accomplishes. The common guillemot, *Uria aalge*, a seabird belonging to the auk family, nests in coastal areas but hunts several kilometres out to sea depending on prey locations, flying as far as 30 km in some instances [13]. Once at the hunting ground the birds dive underwater, morphing their wings, and utilize the same flapping mechanism in water as they do in air. Other kinematic changes also take place, such as the flapping frequency reducing from approximately 9 to 2.5 Hz when moving from air to water, and the forward velocities reduce from 19.1 to 1.52 m $s^{-1}$, respectively [12, 14]. This alteration in wing shape can be seen in figure 1.

This ability is clearly unmatched within the engineering world and as such offers a challenging task in trying to reproduce this technique with any degree of success. In this investigation, the savings in profile drag relating to the retracted wing during aquatic operations are empirically determined. A numerical model is then generated to try and capture key factors the guillemot and other similar bird species come up against. As highlighted in the introduction, current limitations of both understanding and technologies mean that performing this task can be difficult, especially considering the two different mediums.

## 3. Power requirements of the multi-modal common guillemot

### *3.1. Initial theoretical work*

During flight, the guillemot must overcome external resistance in order to maintain steady horizontal flight. By determining

these, the overall power requirement for flight can be calculated. These forces can be broken down into three distinct forms of drag that the bird is subjected to [15].

The first component is lift-induced drag, calculated by using the following equation:

$$\text{Drag}_{\text{Ind}} = \frac{2k_{\text{induced}}m^2g^2}{\pi\rho_{\text{air}}U^2b^2}, \quad (1)$$

where $k_{\text{induced}}$ is the induced drag factor, $m$ is the mass of the bird (kg), $g$ is the gravitational constant, $U$ is the air velocity (m s$^{-1}$), $\rho_{\text{air}}$ is the density of air (kg m$^{-3}$) and $b$ is the wing span (m). The second form of drag is that of parasitic drag, often called body drag, which is made up of skin friction and pressure drag associated with the main body section of the bird. This is calculated using the following equation:

$$\text{Drag}_{\text{Par}} = 1/2\rho_{\text{air}}S_bU^2C_{\text{D-Bod}}, \quad (2)$$

where $S_b$ is the frontal area of the body (m$^2$) and $C_{\text{D-Bod}}$ is empirically determined values of the body drag coefficient. Previous experimental work has empirically determined parasitic drag values for a number of bird species, including the common guillemot, *Uria aalge*, which allows the use of equation (2) by providing values for $C_{\text{D-Bod}}$ [16].

The final component is the profile drag of the airfoil, made up of skin friction and pressure drag. Both contributions can be theoretically calculated but to do this knowledge of the pressure distribution around the wing is required and also the shear stress distribution on the surface. Determining these can be a very laborious process therefore when required accurately the profile drag is typically measured experimentally. However profile drag can be calculated by the following equation:

$$\text{Drag}_{\text{Pro}} = 1/2\rho_{\text{air}}SU^2C_{\text{D-Pro}}, \quad (3)$$

where $S$ is the wing area (m$^2$) and $C_{\text{D-Pro}}$ is the profile drag coefficient, using empirically determined values for $C_{\text{D-Pro}}$. The lack of experimental work on guillemot wings resulted in an uncertainty in profile drag coefficients. It was therefore decided that an experimental procedure should be conducted in order to establish typical values.

Having now isolated each type of drag the guillemot will be subjected to and by utilizing the fact that power is the product of force and velocity and by multiplying each by the mean airspeed, the power associated with each drag force can be calculated. Combining these gives an estimation of the overall aerial power requirement:

$$P_{\text{Ind}} = 2\ \text{km}^2\text{g}^2/\pi\rho_{\text{air}}Ub^2 \quad (4)$$

$$P_{\text{Pro}} = 1/2\rho_{\text{air}}SU^3C_{\text{D-Pro}} \quad (5)$$

$$P_{\text{Par}} = 1/2\rho_{\text{air}}S_bU^3C_{\text{D-Par}} \quad (6)$$

$$P_{\text{Sum}} = P_{\text{Ind}} + P_{\text{Pro}} + P_{\text{Par}}. \quad (7)$$

During aquatic locomotion the guillemots again travel within a fluid, but rather than in air as with flight, they have changed medium to water which has a major impact on the power requirements of the locomotion technique. When travelling underwater the guillemot operates in a medium some 800 times denser than air, resulting in the drag associated with both the body and wings being far greater under water at comparable velocities.

The bird is no longer subjected to large forces due to gravity in the vertical direction, whilst submerged and can be considered to be virtually neutrally buoyant. In actual fact the guillemot must counter forces lifting them to the surface due to air in their plumage, but during this research it has been assumed that these forces are negligible.

The drag associated with aquatic locomotion is again profile drag consisting of pressure drag and skin friction drag. Whether pressure drag or skin friction drag dominates depends on the size and shape of the object in question, and as discussed with aerial flight, to determine profile drag with any real accuracy experimental work must be performed.

Similar to aerial operations for guillemots flapping underwater the different drag contributions are the body drag and profile drag, using equations (2) and (3) respectively but with the different value for the density. Total power requirements associated with drag for aquatic locomotion can then be determined utilizing the same relationship as utilized in equations (5) and (6).

In previous work [17] it was proposed that the power requirements of the common guillemot associated with drag during both aerial and aquatic modes of locomotion are likely to be similar. Initially an estimation was made in [17] of the power requirements of the common guillemot when swimming underwater based on a combination of values gathered by experimental work by [16], who examined the drag of the body of the common guillemot, combined with other wing profile drag values based on equation (2). The estimate for aerial power requirements gave an initial value of 16.96 W. The value for the total power required by the bird to overcome frictional forces during aquatic operations was estimated at 17 W; however this value did not take into account the drag due to wings. The experimental work covered in the following work section empirically determines the saving in profile drag associated with the retracted wings during aquatic operations as well as profile drag coefficient values associated with the guillemot wing during both modes of locomotion.

To determine the potential saving associated with the alteration of wing shape between modes and the drag coefficients, static guillemot wing representations have been designed, manufactured and tested in a range of velocities and orientations.

### *3.2. Wing models*

The static guillemot wing models were developed using a solid modeller (NX3, Siemens). Through a combination of available literature and images, wing models were developed to represent the wing shape during both aerial and aquatic modes of locomotion. These models encapsulated a typical wing size based on a selection of literature [14, 15; 18, 19] and can be seen in figures 2 and 3. Utilizing the length between the ulna/metacarpals joint and the wing tip, a length which is constant in both aerial and aquatic models, corresponding models for both mediums based on the same geometry were created.

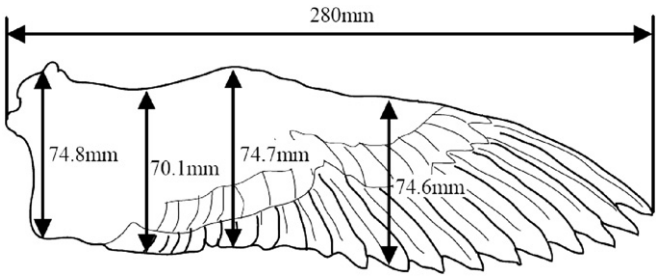



**Figure 2.** Wing dimensions determined for use in solid model aerial design (adapted from [18]).

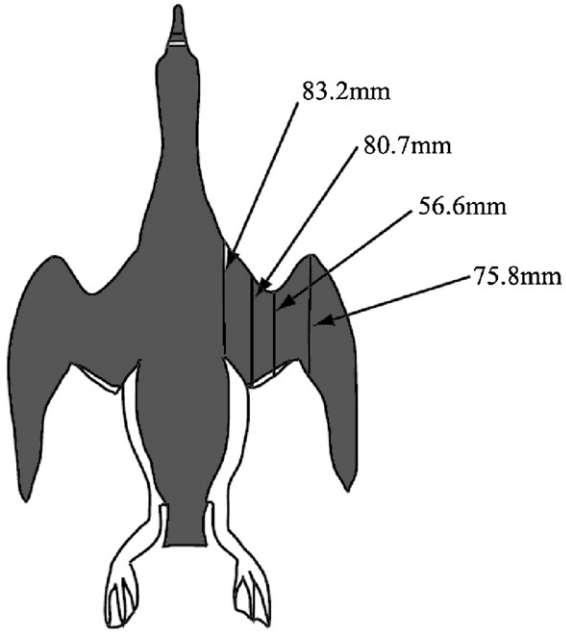



**Figure 3.** Chord lengths used in a aquatic wing model [19].

The wing models were subsequently manufactured using a rapid prototyping machine at the Bristol Robotics Laboratory (BRL), using a FDM Titan (Stratasys). The chosen material was Polycarbonate-ISO with the models being created using a layer thickness of 0.007 inch (0.178 mm). The models were painted flat black to assist in the visualization of the wing models during testing.

### *3.3. Experimental method*

Testing was carried out in the Hydrodynamics laboratory in the Department of Civil Engineering at the University of Bristol. The water flume used had an overall length of 10 m, a width of 0.3 m and a maximum possible depth of 0.5 m with the water depth kept at a constant level of 0.325 m. The flume can be seen in figure 4.

The drag of the wings was calculated by controlling the horizontal velocity of the apparatus, whilst continuously recording the associated drag. The driving system, shown in figure 5, was connected by chord to a movable platform able to travel along horizontal tracks attached to the upper edges of the flume. The wing models were suspended from the platform, which when mounted to the flume submerged the wings within the water, demonstrated with the aquatic wing model attached in figure 6.



**Figure 4.** Flume used during experimental testing.



**Figure 5.** Winch chord continuously winding around the drum.

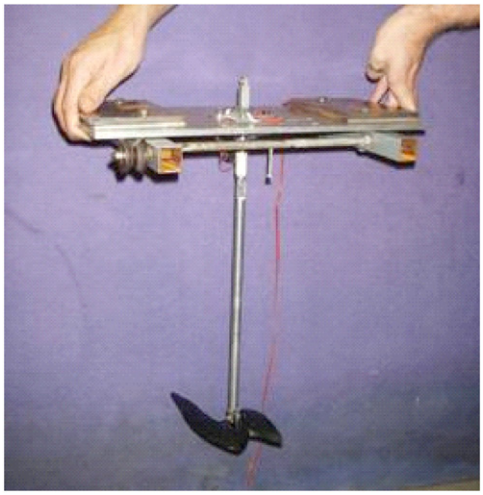


**Figure 6.** Overall platform assembly.

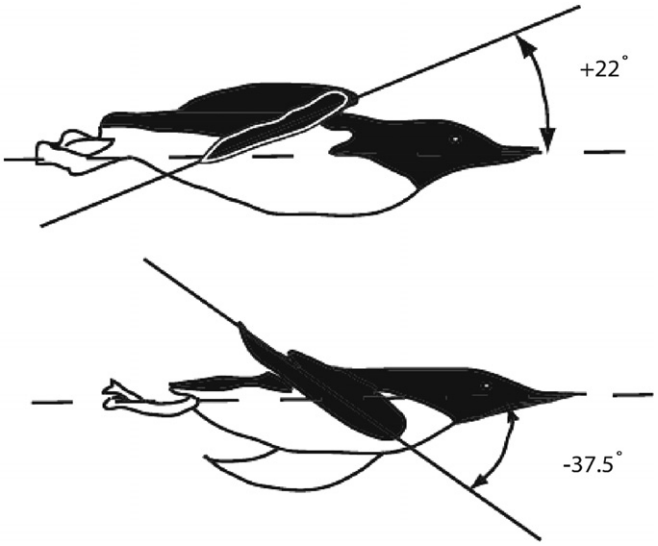



**Figure 7.** Maximum and minimum pitch angles of a wing during aquatic locomotion (adapted from [19]).

Strain gauges, type GFLA-3-70 (Tokyo Sokki Kenkyujo Co., Ltd), were attached to the support beam to measure the tensile and compressive strains experienced in the beam which were converted to equivalent horizontal drag following initial calibration. The readings were fed through a strain indicator, type 1526 (Bruel & Kjaer Ltd). To allow continuous readings to be recorded the signal from the strain indicator was converted to a digital signal. To achieve this dSpace was used in conjunction with a model created in Simulink, part of the software package Matlab (Mathworks, Inc.), which was subsequently generated in the ControlDesk environment of the dSpace software, acting as the real-time testing environment for the extrapolation of data.

A Model 400 B2 control unit (Neco Electronics Ltd) was used to control the speed of the system, a type BD 1306 dc motor (British Thomson-Houston) was used to power the pulley system and a type 18/175 dc motor (Rank Pullin Control) generated the output voltage based on the RPM of a bespoke drum located at the centre of the pulley system. This enabled simultaneous recordings of forward velocity and the drag experienced by the wing models. The output voltage from the dc motor relating to specific RPMs was determined, which was subsequently converted into an equivalent horizontal velocity based on the known drum diameter.

The output voltage was channelled through dSpace. The appropriate gain values were introduced within the ControlDesk environment so that the output signal was automatically converted into the equivalent horizontal velocity of the assembly.

Due to the static nature of the wings, the testing was repeated several times, manually adjusting the orientation of the wings to determine the implications of pitch variation on the drag experienced. Figure 7, taken from [14], shows that during aquatic locomotion the angle of attack of the wings approximately varies between 22° and −37.5° to the horizontal, which acted as the range of pitch motions during investigations. Testing was also conducted over a range of speeds. The tests were performed on both models and repeated four times in each specific orientation.

**Table 1.** Equations relating drag to velocity for both wing models at all tested orientations.

| Wing model/orientation | Equation |
|---|---|
| Aerial: horizontal | $D = 1.9087U^{1.8656}$ |
| Aerial: 5° yaw | $D = 1.9622U^{1.861}$ |
| Aerial: 10° yaw | $D = 1.8774U^{1.9164}$ |
| Aerial: 11° pitch | $D = 2.3479U^{1.6769}$ |
| Aerial: 22° pitch | $D = 4.36U^{1.771}$ |
| Aerial: −19° pitch | $D = 4.0057U^{1.7008}$ |
| Aerial: −37.3° pitch | $D = 7.0658U^{1.6386}$ |
| Aquatic: horizontal | $D = 1.4163U^{1.9233}$ |
| Aquatic: 5° yaw | $D = 1.372U^{1.8301}$ |
| Aquatic: 10° yaw | $D = 1.3091U^{1.7844}$ |
| Aquatic: 11° pitch | $D = 1.5318U^{1.8358}$ |
| Aquatic: 22° pitch | $D = 2.2774U^{1.8152}$ |
| Aquatic: −19° pitch | $D = 2.1416U^{1.7393}$ |
| Aquatic: −37.5° pitch | $D = 3.626U^{1.7963}$ |

### *3.4. Experimental results*

For both the aerial and aquatic models, experimental runs were conducted at each of the discretized values of pitch angle, being towed at forward velocities ranging from 0.5 to 2.5 m s$^{-1}$. The overall data sets relating to each model were analysed in order to determine trend lines for each model and orientation which can be found in table 1.

By utilizing the equations in table 1 and using increments of 0.25 m s$^{-1}$, the drag associated with each wing model could be calculated for the range of pitch angles at various forward velocities. These can be seen in figures 8 and 9. Polynomial trend lines that provided an accurate representation of the plots were determined for each data series. The averages of these functions were then calculated for each speed increment by integration of the polynomial trend lines between the maximum and minimum pitch angles. These average values, along with trend lines are shown in figure 10.

Using the equations of the trend lines the average profile drag experienced by each wing can be determined for any particular speed of interest. By utilizing these equations and using the average speed of 1.52 m s$^{-1}$ [12], the aquatic wing model gives an average value of 3.03 N and the aerial wing model gives a value of 6.12 N. This is based on the assumption that the guillemot flaps its wings to the maximum and minimum amplitude during locomotion, i.e. a maximum possible saving This demonstrates that by utilizing the adjusted wing shape in water the guillemot can reduce the wing profile drag by up to 50.5%.

The drag coefficients of each wing model can also be found from the gathered experimental results. By then plotting these values against Reynolds numbers a dimensionless relationship is developed which can be used in subsequent analytical processes.

The Reynolds number for each wing model can be calculated for the different speed increments by using the

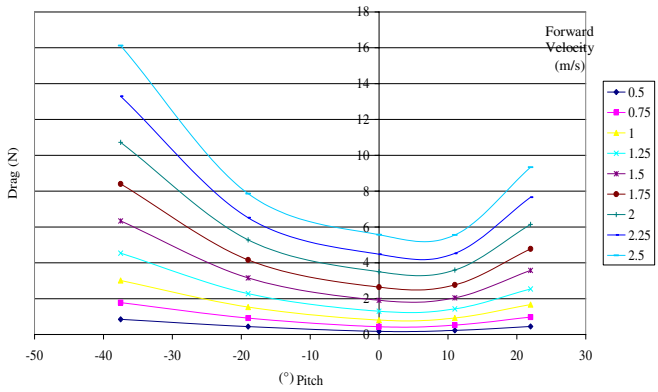



**Figure 8.** Graph to show drag versus pitch angle for various speeds—Aquatic wing model.

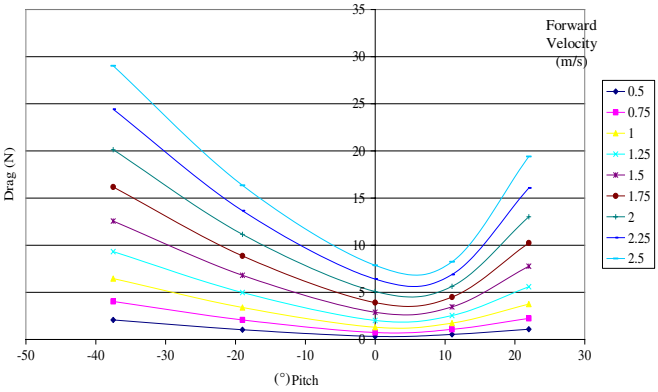



**Figure 9.** Graph to show drag versus pitch angle for various speeds—Aerial wing model.

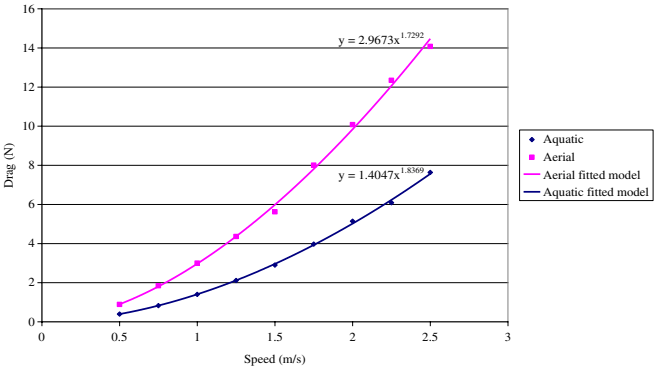



**Figure 10.** Graph to show average drag for complete wing beat versus speed for both wing models, inc. trend lines.

following formula:

$$Re = \frac{lU}{\upsilon}, \tag{8}$$

where $l$ is the characteristic length (m), $U$ is the speed (m s$^{-1}$) and $\upsilon$ is the kinematic viscosity of water ($1.141 \times 10^{-6}$ m$^2$ s$^{-1}$ at 15 °C). The characteristic length is typically taken as the average chord length of the wing when dealing with foils. These lengths have been calculated as 0.070 12 m and 0.085 05 m for the aerial and wing model, respectively.

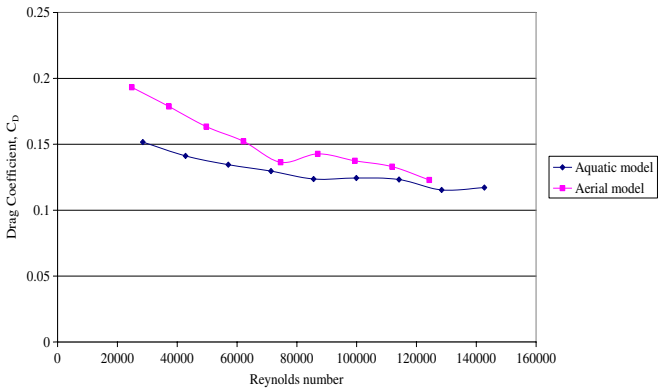



**Figure 11.** Drag coefficients, $C_D$, versus Reynolds number for the static wing models.

The formula for profile drag can be rearranged to give the drag coefficient at specific velocities as follows:

$$C_{\text{D-Pro}} = \frac{2 \times D_{\text{Pro}}}{\rho_{\text{water}} S U^2}, \tag{9}$$

where $D_{\text{Pro}}$ is the profile drag (N). Therefore by using equations (13) and (14) and the average values of profile drag as calculated by the trend lines in figure 10, plots of $C_{\text{D-Wing}}$ versus Reynolds number are created for both wing models, shown in figure 11. It can be seen that the aquatic wing model has a lower drag coefficient over the entire range of testing.

### *3.5. Experimental discussion*

It must again be stressed that the empirical studies main aim was showing that the guillemot has optimized the wing orientation based on the characteristics of the two very different mediums. There are obvious limitations in the level of mimicry and analysis that can be achieved by looking at solely static wing models. However by analysing the empirical results there is no doubt that by retracting the wing during aquatic locomotion the profile drag is reduced, reinforced by the fact that the drag coefficient of the aquatic wing model remains less than that of the aerial wing model over the range of tested Reynolds numbers.

Analysing figures 9 and 10 relating to the aquatic and aerial models respectively shows that once the wing is pitched beyond thresholds in both positive and negative directions for the aquatic model the drag experienced rises sharply above +10° and −20°, also observed for the aerial model again above +10° but less so in the negative orientation. To account for this the dominating type of profile drag involved with the various orientations needs to be considered. Take the common example of a small rectangular plate in a fluid flow; when held in the fluid 'edge on' (i.e. wing in the horizontal position), predominant profile drag is due to the skin friction component. However when the plate is held perpendicular to the flow (i.e. similar with wing models when pitch variation is large), the predominant profile drag is due to pressure drag. This variation in pitch greatly alters the profile of the wings, resulting in high drag at large pitch angles and can account for the flat sections

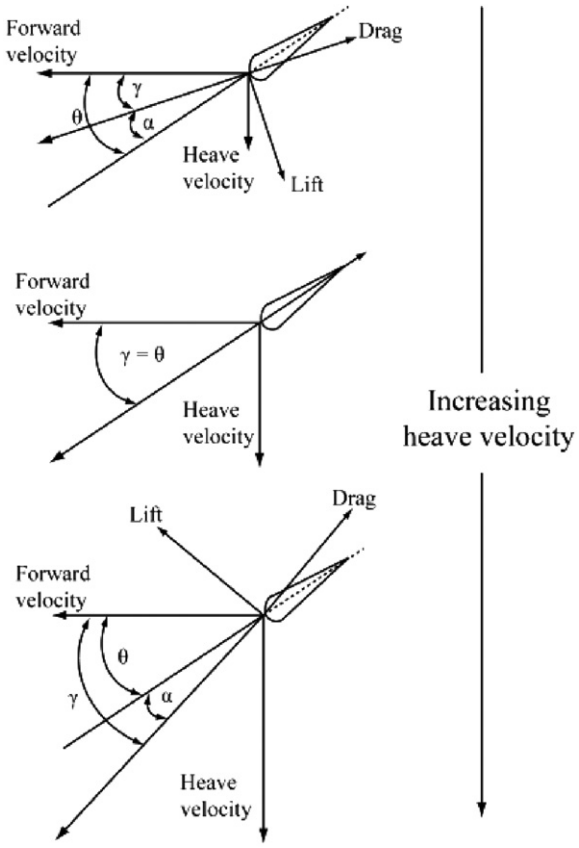



**Figure 12.** Implications of increasing heave velocity on localized wing kinematics.

within figure 8, where profile drag is dominated by skin friction drag rather than higher pressure drag.

There was however a limitation in the experimental procedure in that the oncoming fluid flow was continuously horizontal for the entire wing. This is in fact not the case, with the wing during flapping experiencing both a horizontal and vertical velocity component based on the forward motion and the vertical component from the flapping action. The implications of this on the overall power requirements of the guillemot are covered in more detail in the following numerical section; however, figure 12 demonstrates what occurs at various points spanwise along the wing.

The heave velocity associated with the wing greatly affects the fluid dynamics associated with different parts of the wing. Referring to figure 12 it can be seen that once the pitch angle, $\theta$, exceeds the fluid approach angle, $\gamma$, the wing would in fact be creating lift in the negative thrust direction at this section of the wing. This is obviously not desirable. This behaviour is dealt with in engineering applications by employing a twist of the foil, i.e. in a propeller, so that angle of attack of the blade varies spanwise. Unfortunately, the authors are unable to confirm whether the guillemot's wing twists during aerial or aquatic locomotion.

Relating back to figure 7 looking at the maximum and minimum pitching angles of the aquatic wing, it can be seen that the maximum pitch amplitude is greater on the downstroke than the upstroke. By analysing video footage the authors have identified that a guillemot flapping at approximately 2 Hz spent 0.22 ($\pm$0.02) s on the downstroke and 0.3 ($\pm$0.02) s on the upstroke, implying a faster downstroke velocity. Looking at figure 12 this implies that with a greater heave velocity on the downstroke, the actual pitching angle of the wing can be larger than on the upstroke without any negative effects on thrust production, highlighting the fact that the guillemot has optimized the kinematics of the wing to maximize thrust production.

This relationship has been highlighted for a crucial reason. Considering the localized kinematics of the wing, i.e. the heave and forward velocity, the actual experienced angle of attack ($\alpha$) will be far less than the value the static wing model was subjected too. The potential implications of this are that the actual typical profile drag experienced by the wing, in both orientations, is less than the values given.

It has been shown that the drag experienced by the aquatic model is approximately half of that experienced by the aerial model for the entire speed range, not just at 1.52 m $s^{-1}$. It is therefore conclusive that the folding arrangement of the wing is advantageous over a range of velocities in reducing the associated profile drag.

The calculated overall power requirement associated with body drag for aquatic locomotion based on values measured during this research and results from body drag calculated in [16] gave a total power requirement of 12.0 W at a speed of 1.52 m $s^{-1}$. By comparing this to the power requirement for flight, calculated as 16.96 W from first-order calculations in [17], as would be expected they are similar orders of magnitude. Considering the likelihood that the empirically determined drag values are potentially higher than those given once considering the full kinematics, this would again reduce the aquatic power requirements associated with drag, however still remaining the same order of magnitude. However, additional factors must also be considered when calculating aquatic overall power requirements.

Flapping motions in the submarine environment must also contend with the inertial requirement of the added mass associated with the water. In air this is not such an issue at this scale of flapping [20], but as the density of water is approximately 800 times greater, the additional power required to set this mass of water in motion must be accounted for. Analysis at this stage still indicates that the folding arrangement is very beneficial in reducing profile drag, however the development of a numerical model that accounts for the dynamic motion of the guillemot wing is now created in an attempt to quantify the implications on power requirements when flapping the full and retracted wings underwater taking into consideration the varying kinematics along the wing and introducing inertia contributions into the calculations of power requirements.

### *3.6. Numerical model development of a multi-modal wing*

The developed model has three key purposes: as a means of furthering our understanding of the dynamics associated with the flapping motion during aquatic locomotion, as a means to investigate the compromises made by the guillemot by using the same propulsion mechanism in two different mediums and also as a potential future design tool for vehicles of this type.

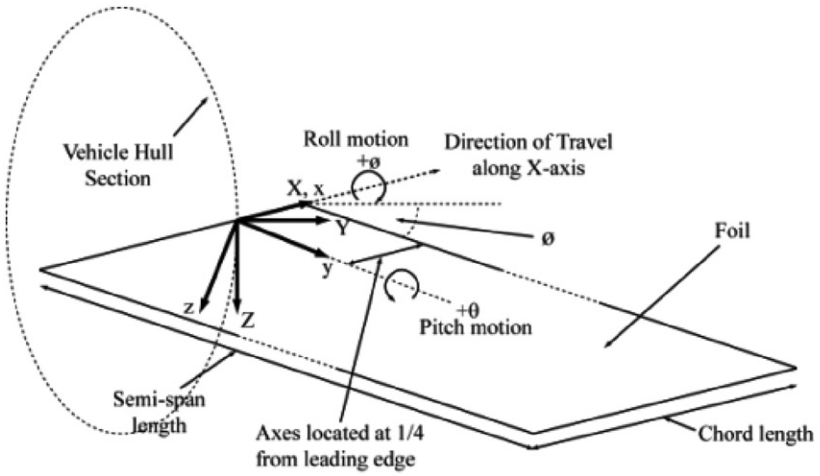



**Figure 13.** Body-fixed and wing-fixed axes positions with a retraction angle, $\beta = 0$.

For simplicity the wing has been modelled as a flat plate, being retracted backwards forming a parallelogram. Although the guillemot wing retracts in a more complex manner, the authors believe this simplified model will still capture and help demonstrate implications of reducing wing plan area in terms of overall thrust production and power requirements, introducing the implications of inertial dynamics experienced by the flapping motion. It will not however be hydrodynamically equivalent to the actual guillemot wing but acts as a next step in furthering our understanding of the processes involved in this form of multi-modal locomotion.

*3.6.1. Aquatic operations.* In the aquatic model the key focus is on the behaviour of the flapping foil alone. A simplified representation of the foil orientation is therefore proposed, which captures the main points of interest. Figure 13 displays the axes and subsequent sign conventions that accompany the foil model. The axes orientations are chosen to be in keeping with the standard format used in AUV design [21]. The folding mechanism is a single folding four-bar mechanism, simplifying the retraction technique of the guillemot.

In addition to the body-fixed axes, *XYZ*, an additional set of axes are attached to the flapping wing, namely *xyz*, coincident with the body-fixed axes when the roll, pitch and retraction angle of the foil are zero. These axes are considered to rotate with the wing during roll motion. The foil is considered to be joined to a hypothetical body section represented by the circular dotted line in figure 13 at the same point at which the axes *XYZ* and *xyz* intersect when $\varphi$ and $\theta$ are both zero.

*Hydrodynamics.* The hydrodynamic model utilized in this simulation is based on quasi-steady blade element theory. This has been confirmed as a suitable method for calculating the forces associated with the flapping motion in previous modelling of the Humboldt penguin [10]. For each geometric arrangement the wing is split into finite segments in the spanwise direction which are analysed independently based on the local conditions. The foil is assumed rigid throughout the course of analysis.

In order to be able to utilize blade element theory (BET), fully described kinematics must be known for the wing throughout the wing beat cycle. Several birds utilize a flapping technique underwater that develops lift on both the up and down strokes, which is then resolved into forward thrust, including the common guillemot [11]. The kinematic trajectory of the foil was chosen to be similar to this, utilizing a two-degree-of-freedom flapping motion consisting of roll and pitch operating in a harmonic motion with the same circular frequency $\omega$ (rad s$^{-1}$). Firstly, the roll motion representing the wingbeat amplitude is governed by

$$\phi(t) = \phi_0 \sin(\omega t), \tag{10}$$

where $\phi 0$ is the maximum amplitude in radians. Similarly the pitch motion, representing the harmonic pronation and supination of the wing, is governed by

$$\theta(t) = \theta_0 \sin(\omega t + \psi), \tag{11}$$

where $\theta 0$ is the maximum pitch amplitude in radians and $\psi$ is the phase angle between pitch and roll. It has been highlighted in previous work that a phase lag of 90° is the most suitable arrangement which shall be adhered to here [22]. This results in equation (2) becoming

$$\theta(t) = \theta_0 \cos(\omega t). \tag{12}$$

Provided the maximum wingbeat amplitude and pronation/supination amplitudes are known, the kinematics of the flapping foil are fully described. This is graphically shown in figure 14.

Previous research into flapping foils utilized in AUV designs has shown that each blade element can be considered to be moving in a heave like motion [23]. Subsequently if the forward velocity that the foil is travelling at is known and the heave velocity is calculated for each element, the fluid approach angle, $\gamma$, can be calculated based on these values, as shown in figure 12.

In addition to the fluid approach angle, the overall foil continuously changes the pitch angle throughout the course

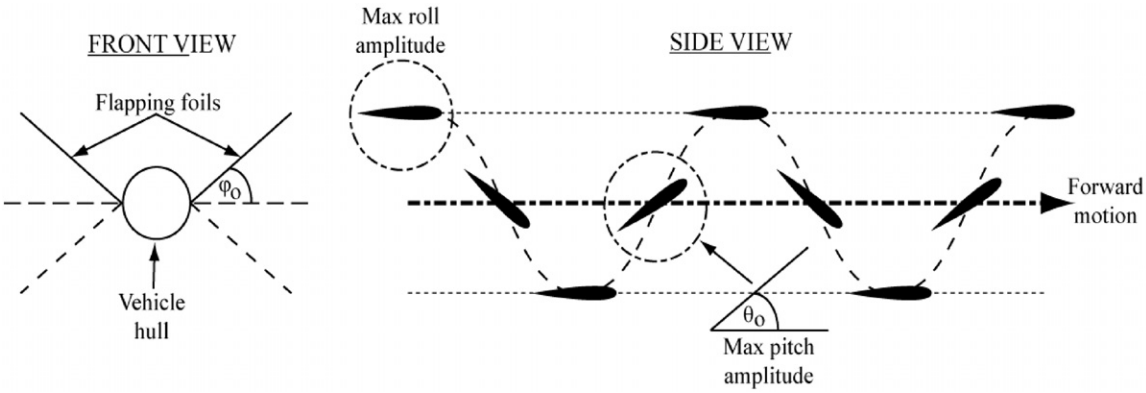



**Figure 14.** Graphical representation of harmonic nature of the roll and pitch motions, demonstrating the phase lag of maximum roll and pitch amplitudes.

of the wing beat. Therefore this kinematic parameter is also required so as to determine the effective angle of attack that each blade element experiences. This can again be seen in figure 12.

The chosen kinematics of the pitching motion therefore plays a crucial role in the overall foils ability to generate forward thrust. For every blade element, each with its own kinematic characteristic, the lift and drag are determined by the following:

$$\mathrm{d}L = 0.5\rho V_{\mathrm{res}}^2 \mathrm{d}SC_l, \tag{13}$$

$$\mathrm{d}D = 0.5\rho V_{\mathrm{res}}^2 \mathrm{d}SC_d, \tag{14}$$

where d$L$ is the lift associated with a specific element, d$D$ is the drag associated with the same specific element, $\rho$ is the density of the water, *V*res is the resultant velocity due to both the forward velocity and heave velocity, d$S$ is the characteristic area and *Cl* and *Cd* are the lift and drag coefficients respectively based on the specific foil profile.

The lift coefficient and the drag coefficient will vary according to the effective angle of attack that each element experiences due to the localized conditions; therefore a relationship between the angle of attack and the lift and drag coefficients must be determined. To do this the lift and drag coefficient profiles are mapped to a harmonic profile as shown in figure 15 [24]. This means that the lift and drag coefficient can be calculated for any angle of attack provided the maximum values of lift and drag coefficients are known. Equations (4) and (5) subsequently become

$$\mathrm{d}L = 0.5\rho V_{\mathrm{res}}^2 \mathrm{d}SC_{L\,\max} \sin(2\alpha) \tag{15}$$

$$\mathrm{d}D = 0.5\rho V_{\mathrm{res}}^2 \mathrm{d}SC_{D\,\max}(1 - \cos(2\alpha)). \tag{16}$$

Although the guillemot has a cambered wing profile typical of avian species, during the numerical modelling a symmetrical foil profile has been used, specifically an NACA0012 profile [25]. This provides a symmetrical relationship relating lift and drag coefficients in both positive and negative pitching. The reduction in the maximum lift coefficient value compared with cambered high lift producing profiles was considered acceptable given chosen harmonic flapping in water.

For each element the drag acts in the same direction as the fluid approach angle, with the lift acting perpendicular to this. As the foil model is considered to be moving along the *X*-axis, the lift and drag components of each element along the *X*-axis can be determined by a trigonometric function. This value along the *X*-axis represents the thrust generated by the flapping foil, one of the performance measures considered in the following sections. Similar transformations are done for the *Y*- and *Z*-directions. However, in addition to resolving the forces in relation to the fluid approach angle, they must again be resolved based on the position due to the roll motion. Each blade element component is calculated by the following:

$$\mathrm{d}F_X = \mathrm{d}L(\sin(\gamma(t))) - \mathrm{d}D(\cos(\gamma(t))) \tag{17}$$

$$\mathrm{d}F_Y = (\mathrm{d}L(\cos(\gamma(t))) + \mathrm{d}D(\sin(\gamma(t)))).\sin(\phi(t)) \tag{18}$$

$$\mathrm{d}F_Z = (\mathrm{d}L(\cos(\gamma(t))) + \mathrm{d}D(\sin(\gamma(t)))).\cos(\phi(t)), \tag{19}$$

where d$FX$, d$Fy$ and d$Fz$ are the resolved forces in the *X*-, *Y*- and *Z*-axes, d$L$ and d$D$ are the lift and drag respectively for the element in question, $\gamma(t)$ is the fluid approach angle and $\phi(t)$ is the roll angle of the foil.

The instantaneous forces along all the body-fixed axes for the entire wing can therefore be determined by combining these values for each blade element.

Each blade's hydrodynamic moment can also be calculated about the body-fixed axes based on the localized forces experienced at each element and the distance from the point of rotation. Similar to the forces experienced about specific axes, the overall hydrodynamic moments experienced about the body-fixed axes can be calculated by summating the calculated values over the semi-span length of the wing.

In addition to the forces and moments acting about the body-fixed *XYZ* axes, the forces must also be resolved about the wing-fixed axes, *xyz*. This is again performed by means of a rotational transformation.

*Inertial dynamics.* Inertial dynamics must also be considered to be able to determine the torques required to drive the flapping motion, which subsequently lead to power requirements. In order to develop a generic inertia tensor that could be utilized for the various geometric arrangements the wing was assumed to be the equivalent of a specific thickness

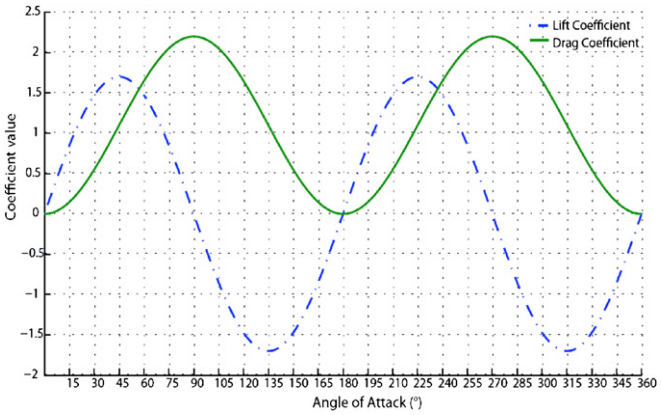



**Figure 15.** Lift and drag coefficients versus angle of attack.

flat plate. The generic inertia tensor was determined for the flat plate based on the geometric variables shown in figure 5 subsequently referred to as $I_{\mathrm{RB}}$.

In addition to the inertia associated with the rigid body of the wing, consideration must also be given to the inertia associated with the surrounding fluid set in motion due to the flapping foil. It has been shown by inviscid flow theory that the added mass of a foil accelerated in a perpendicular direction to its chord is equal to an imaginary circular cylinder encapsulating the wing, with the chord as its diameter [20]. The foil therefore can be assumed to have an added mass per unit span equal to

$$m_a = 1/4\rho_{\mathrm{water}}\pi c^2, \tag{20}$$

where $c$ is the chord width of the foil. To calculate the overall inertia tensor associated with the added mass, the foil was again considered as finite strips analysed individually. Each strip was then assigned an equivalent cylindrical mass based on the chord width as described by equation (18). This was then treated as a point mass located at the centre of the foil and placed at the equivalent half chord location of each segment. The moments and products of inertia associated with each segment were determined in relation to the body-fixed axes. As the moments and products of inertia of each point mass are calculated with respect to the wing-fixed axes, these can be combined to form the overall moments and products of inertia for the entire added mass associated with the wing. This overall value represents $I_A$ for any foil geometry arrangement. Furthermore, as $I_{\mathrm{RB}}$ and $I_A$ are calculated about the same axes the values can be summated to provide the overall inertia tensor to be used when calculating the inertial dynamics of the system.

The angular velocities and accelerations of the wing can be calculated by considering the Euler angles due to the three rotations: pitch, roll and retraction angle. Firstly, a rotation is experienced about the $X$ axis corresponding to the roll of the wing, secondly, a rotation due to the pitching moment and finally, the rotation due to the retraction angle $\beta$, providing an overall transformation represented by the following:

$$[R] = [R_\beta][R_\theta][R_\phi]. \tag{21}$$

Considering all three rotations the overall angular velocity can be calculated based on the vector sum of the individual rotation rates, leading to

$$\underline{\omega} = \dot{\beta}\,\underline{k} + \dot{\theta}\,\underline{j}' + \dot{\phi}\,\underline{\mathrm{I}}, \tag{22}$$

which simplifies to

$$\underline{\omega} = \dot{\theta}\,\underline{j}' + \dot{\phi}\,\underline{\mathrm{I}} \tag{23}$$

as the retraction angle $\beta$ is constant during each geometric arrangement being analysed.

The angular velocities and angular acceleration about the wing-fixed $xyz$ axes are subsequently determined as follows:

$$\underline{\omega} = (\dot{\phi}\cos\beta\cos\theta - \dot{\theta}\sin\beta)\underline{i} + (\dot{\theta}\cos\beta - \dot{\phi}\sin\beta\cos\theta)\,\underline{j} + (\dot{\phi}\sin\theta)\underline{k} \tag{24}$$

$$\underline{\alpha} = (\ddot{\phi}\cos\beta\cos\theta - \ddot{\theta}\sin\beta - \dot{\theta}\dot{\phi}\sin\theta\cos\beta)\underline{i} + (\ddot{\theta}\cos\beta - \dot{\theta}\dot{\phi}\sin\theta\sin\beta - \ddot{\phi}\sin\beta\cos\theta)\,\underline{j} + (\ddot{\phi}\sin\theta + \dot{\theta}\dot{\phi}\cos\theta(\cos^2\beta - \sin^2\beta))\underline{k}. \tag{25}$$

Utilizing the inverse of the rotation matrix governed by equation (19) the angular velocities and angular accelerations about the body-fixed $XYZ$ axes can be calculated:

$$\underline{\omega}_{XYZ} = [R]^T\,\underline{\omega}_{xyz} \tag{26}$$

$$\underline{\alpha}_{XYZ} = [R]^T\,\underline{\alpha}_{xyz}. \tag{27}$$

The torque experienced about the $xyz$ axes can be calculated based on the rate of change of angular momentum, where angular momentum is expressed by

$$\underline{H}_{xyz} = I_{xyz}\underline{\omega}_{xyz}. \tag{28}$$

As the wing is assumed rigid, the inertia tensor remains constant, so the torque experienced about the wing-fixed $xyz$ axes is

$$\underline{\tau}_{xyz} = \dot{\underline{H}}_{xyz} = I_{xyz}\underline{\alpha}_{xyz}. \tag{29}$$

A similar process is used to determine the torques experienced about the body-fixed $XYZ$ axes with one slight difference. The calculated inertia tensor is based on the wing-fixed axes location. Subsequently, the inertia tensor associated with the body-fixed $XYZ$ axes changes over time.

As the inertia tensor is a tensor of the second rank, a similarity transformation must be used when dealing with rotations. Also, as the rotation matrix, given by equation (15) previously, relates the $XYZ$ axes to the $xyz$ axes, the inverse of the transformation must be used to transform the inertia tensor about the $xyz$ axes to that about the $XYZ$ axes. This is done by the following:

$$I_{XYZ} = [R]^T[I_{xyz}][R]. \tag{30}$$

By combining equations (25) and (28) the torques experienced about the body-fixed $XYZ$ axes can be calculated by

$$\tau_{XYZ} = \dot{\underline{H}}_{xyz} = [R]^T[I_{xyz}][R]\cdot[R]^T\,\underline{\alpha}_{xyz}. \tag{31}$$

Having calculated the hydrodynamic forces and the rigid body inertial contributions relating to the motion of the wing a simplified equation of motion can be created to represent the overall system. The generic dynamic equations of motion can be expressed as

$$M\dot{\upsilon} + C(\upsilon)\upsilon + D(\upsilon)\upsilon + g(\eta) = \tau, \tag{32}$$

where $M$ is the inertia matrix, including the added mass, $C(\upsilon)$ is the matrix of Coriolis and centripetal terms, again including the added mass, $D(\upsilon)$ is the damping matrix, $g(\eta)$ is the vector of gravitational forces and moments and $\tau$ is the vector representing the external torques acting on the systems [21].

Several assumptions are made in order to simplify this into a form that can be utilized in the numerical simulations based on the hydrodynamic and inertial dynamic contributions covered above.

The assumption is made that the wing is only undergoing angular accelerations due to the pitch and flapping motions, and any contributions due to the rotations that would contribute to the overall linear velocities and accelerations are considered negligible at this stage of analyses, meaning there is no linear acceleration experienced by the wing and that all inertial components come from rotational rather than translational motions.

Although present in the real system the Coriolis and centripetal term and the damping matrix shall be left out of the equation of motion in order to simplify the system model. Finally, the wing is assumed neutrally buoyant which subsequently allows the assumption that the gravitational terms can be disregarded during the analysis.

The simplified equation of motion about the wing-fixed axes $xyz$ can therefore be represented by

$$I_{xyz}\underline{\alpha}_{xyz} = \tau_{\text{motor_}y} + \tau_{\text{hydro-}xyz}, \tag{33}$$

where $\tau_{\text{motor-}y}$ is the motor torque required for the pitching motion and $\tau_{\text{hydro-xyz}}$ is the hydrodynamic torque contribution, with the same being true for the roll motion about the body-fixed $XYZ$ axes:

$$I_{XYZ}\underline{\alpha}_{XYZ} = \tau_{\text{motor_}X} + \tau_{\text{hydro-}XYZ}, \tag{34}$$

where $\tau_{\text{motor-}X}$ is the motor torque required for the flapping motion and $\tau_{\text{hydro-}XYZ}$ is the hydrodynamic torque contribution about the body-fixed axes.

Knowing the torques experienced due to the roll motion and pitch motion of the foil and the fact that the angular velocities of both motions are known, an indication of the power requirements required to drive each motion can be determined based on the relationship between power and the torque and angular velocity:

$$P(t) = \tau(t) \cdot \omega(t). \tag{35}$$

Therefore, utilizing the torques calculated above, the power requirements of the driving mechanism behind each motion can be calculated. Subsequently a time-averaged power requirement $\overline{P}_X$ and $\overline{P}_y$ can be deduced for specific geometric arrangements for both the flap and pitch motion.

By combining the time averaged power requirements for both the roll and pitching motions, an indication of the total power required by the system can be determined:

$$\overline{P}_f = \overline{P}_X + \overline{P}_y. \tag{36}$$

The time-averaged total required power into the system to drive both motions can therefore be calculated for any combination of geometric and kinematic parameters.

*3.6.2. Aerial operations.* The aerial model utilizes equations (1)–(6) detailed above, which are traditionally used to determine power requirements associated with avian flight [15].

### *3.7. Model validation*

In order to validate the developed numerical model a hypothetical mission is considered based on the typical behaviour of the common guillemot. From the conclusions of a conducted parametric study [26], assumptions are made within the simulated mission. The span length of the wing is confirmed as the most important geometric variable as this has conflicting benefits on aerial and aquatic performance measures. Chord was found to have a lesser effect than span on power requirements and as such is standardized by considering a wing arrangement with a fixed aspect ratio of 8, based on typical biological examples [13]. The retraction angle, up to the constrained maximum of 45°, was found to be consistently beneficial to the overall power requirements; hence it shall be considered fixed at this value during the aquatic phase.

Knowing the typical ranges of common guillemots when flying out to sea to feed and the subsequent aquatic characteristics whilst hunting as detailed in [12, 13], the aerial and aquatic models can be combined to form an overall scalar cost function relating to the total energy cost of the mission.

An aerial range of 2 km and airspeed of 20 m s$^{-1}$ was selected based on the highlighted literature, resulting in an aerial phase lasting 100 s [12]. An aquatic range of 500 m and aquatic velocity of 1.5 m s$^{-1}$ was selected based on [13], resulting in an aquatic phase ≈333 s. It must be highlighted that this mission represents the bird flying 1 km out to sea, conducting several dives amounting to a total of 500 m over a 333 s time period and then returning to the shore.

During this simulation the key aim is finding the solution set of variables that achieves the highest performance, i.e. the lowest energy cost. Using a continuous search space for several variables results in high computational requirement leading to a large solution space, of which only the best solution is required. As such an alternative optimization strategy needs to be implemented that has a lower computation time. The chosen method uses an evolutionary algorithm to determine the optimal solution set that achieves the highest performance.

Utilizing this process the value with the lowest energy cost was found to be when using a wing of span size 0.74 m with a flapping frequency of 1.76 Hz and a maximum roll and pitch angle of 41.0° and 35.0°, respectively. This is in close agreement with the actual operating conditions of the

guillemot, which has a wing span of approximately 0.71 m and a flapping frequency of 1.9–2.8 Hz, with the pitch angle ranging from +22° to −37.5° as shown in figure 7 [12, 14, 19]. No data were available on the actual flapping amplitude. This therefore helps validate the model and confirms the guillemot behaves with optimal kinematics based on its physiology. A more detailed look at the implications of mission-based parameters and the use of the evolutionary algorithm can be found in [27].

### *3.8. Modelling of wing retraction on aquatic power requirements*

As the aim of this section is to capture and simulate the kinematics and dynamics associated with the flapping motion underwater, a benchmark as a goal for the wing must be set. In [17] it has been shown that at a velocity of 1.52 m $s^{-1}$ the guillemot will experience approximately 2 N of parasitic drag relating to the body. This is therefore set as the target value of thrust production for the different wing orientations. As the bird has two wings this value can be halved as the goal for each wing.

However, due to the fact that there are several kinematic parameters that affect the thrust production these must also be considered. For example, a flapping motion with a small amplitude but with a high frequency can potentially create as much thrust as a flapping motion with a large amplitude but lower flapping frequency. However, the inertial dynamics will differ between the two cases which in turn will have an effect on the power requirements.

In order to identify and compare various combinations of roll amplitude and flapping frequency values for each wing orientation, a continuous search space was initially utilized for each kinematic parameter whilst fixing the wing geometry and forward velocity. A wing span of 0.71 m was selected based on the common guillemot along with the forward velocity of 1.52 m $s^{-1}$, with the folded wing having a retraction angle of 45°. From within this search space for both the extended and retracted wings, solution sets that achieved approximately 1 N of thrust were identified along with the associated moments and power requirements. These results are shown in table 2.

It can be seen from the results in table 2 that the retracted wing provides the required 1 N of thrust at a lesser overall power requirement than the wing when fully extended, true over the range of flapping frequencies and amplitudes. This is the primary conclusion confirming the benefit of folding the wing, but other findings can also be elucidated from the results.

The aim of the numerical model was to investigate the implications of inertial dynamics on the power requirement. It can be seen from table 2 that as the frequency of the flapping increases, so to does the inertial contribution. This can be explained by the fact that the inertial dynamics are strongly related to the angular accelerations of the flapping wing as stated in the numerical model. As the flapping frequency increases, so to does the angular accelerations and decelerations leading to the subsequent increase in inertial contributions, true for both wing orientations.

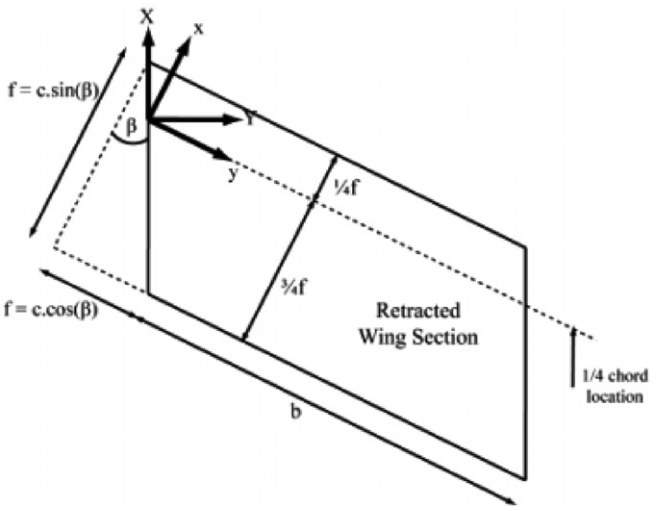



**Figure 16.** Flat plate representation of a foil with an assumed constant thickness.

**Table 2.** Comparison of extended and retracted wings for 1 N thrust generation at varying kinematics.

| $\beta$(°) | Roll (°) | Freq (Hz) | Thrust (N) | Inertial moment (N m) | Hydro moment (N m) | Total moment (N m) | Power (W) |
|---|---|---|---|---|---|---|---|
| 0* | 7.6 | 3.00 | 1.01 | 5.86 | 5.94 | 8.69 | 19.62 |
| 0 | 8.8 | 2.63 | 1.05 | 5.23 | 5.63 | 8.00 | 18.45 |
| 0 | 10.0 | 2.27 | 1.00 | 4.39 | 5.87 | 7.54 | 17.38 |
| 0 | 12.4 | 1.90 | 1.10 | 3.82 | 5.82 | 7.13 | 17.29 |
| 0* | 14.8 | 1.53 | 1.01 | 2.96 | 5.89 | 6.68 | 15.95 |
| 0 | 17.2 | 1.35 | 1.03 | 2.67 | 5.66 | 6.34 | 15.54 |
| 0 | 19.6 | 1.17 | 1.03 | 2.27 | 5.97 | 6.44 | 15.74 |
| 0 | 23.2 | 0.98 | 0.99 | 1.91 | 5.94 | 6.28 | 15.42 |
| 0* | 25.6 | 0.89 | 1.03 | 1.73 | 5.95 | 6.23 | 15.35 |
| 0 | 37.6 | 0.62 | 1.03 | 1.22 | 5.64 | 5.79 | 14.58 |
| 45 | 12.4 | 3.00 | 0.99 | 4.53 | 3.44 | 2.94 | 8.13 |
| 45* | 13.6 | 2.73 | 0.97 | 3.89 | 3.41 | 2.63 | 6.64 |
| 45 | 16.0 | 2.36 | 1.01 | 3.26 | 3.36 | 2.43 | 7.62 |
| 45 | 19.6 | 1.90 | 1.00 | 2.32 | 3.45 | 2.67 | 9.91 |
| 45* | 22.0 | 1.72 | 1.01 | 2.09 | 3.36 | 2.67 | 10.21 |
| 45 | 25.6 | 1.44 | 0.96 | 1.61 | 3.39 | 2.90 | 11.21 |
| 45 | 28.0 | 1.35 | 1.01 | 1.53 | 3.37 | 2.91 | 11.57 |
| 45 | 31.6 | 1.17 | 0.98 | 1.24 | 3.38 | 3.05 | 12.00 |
| 45 | 35.2 | 1.08 | 1.01 | 1.17 | 3.38 | 3.07 | 12.46 |
| 45* | 38.8 | 0.98 | 1.03 | 1.06 | 3.43 | 3.18 | 13.05 |

* Results marked with an asterisk shown graphically in figures 17 and 18.

However what is of most interest is that where as for the extended wing the lowest power requirement is achieved when the inertial contribution is lowest, the lowest power requirement for the retracted wing comes when the inertial contribution is relatively high. In order to explain this, the manner in which the various components of the numerical model vary over time must be considered, with the results marked with an asterisk in table 2 shown in figures 17 and 18.

Firstly, the manner in which the inertial and hydrodynamic moments combine must be considered relating back to the simplified equation of motion given in equation (33). Due to the difference in phase of the different components, the

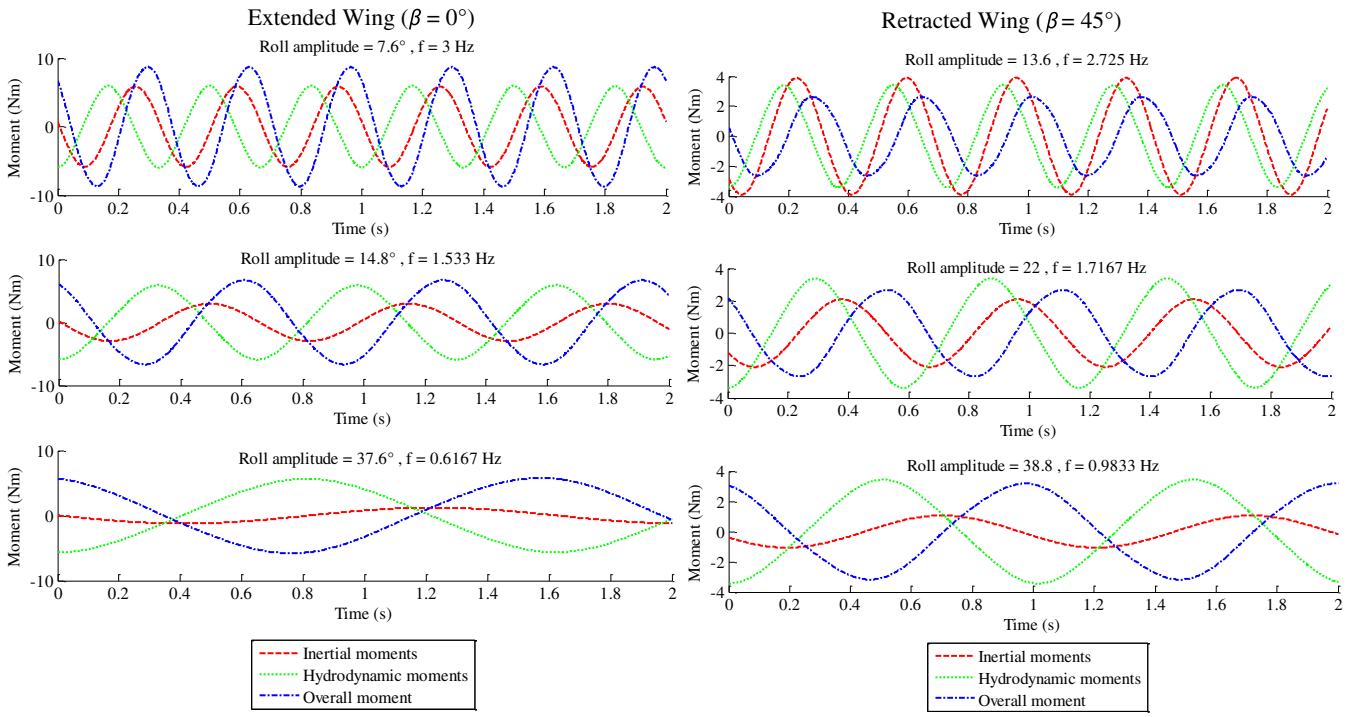



**Figure 17.** Inertial and hydrodynamic moments of extended and retracted wing shapes producing ≈ 1 N of thrust.

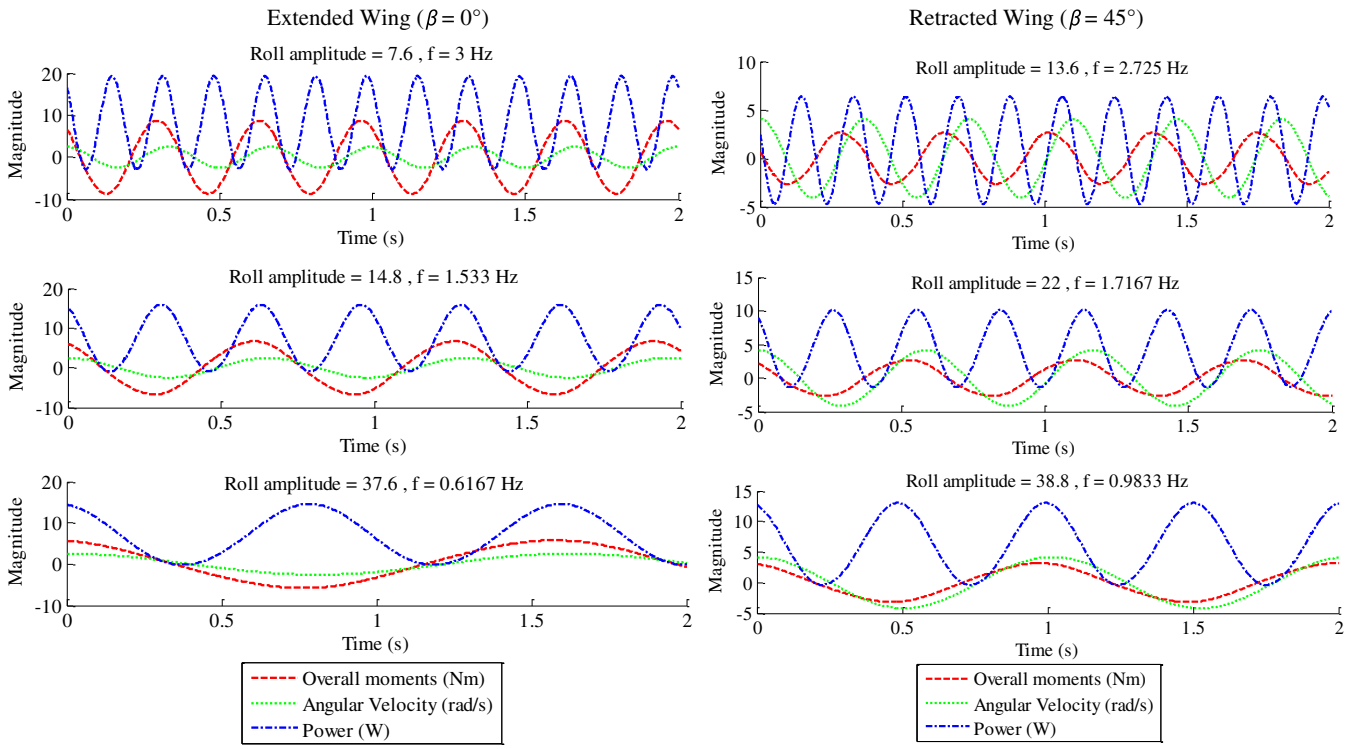



**Figure 18.** Overall moments and power requirements of extended and retracted wing shapes producing ≈ 1 N of thrust.

resultant moment shifts as shown in figure 17. Secondly, the power requirement is calculated by multiplying the resultant moment by the angular velocity, again a time-dependent variable.

The result of combing these components in this manner is that although the wing experiences higher inertial dynamic contributions, the hydrodynamic forces generated and the subsequent hydrodynamic torques on the wing help to alleviate the overall required driving torques. When subsequently multiplying this value by the angular velocity, shown in figure 18, the peak power requirement reduces, with large sections actually implying a negative power requirement.

This is a very interesting finding highlighting even further the importance of optimizing kinematics when flapping underwater, with combinations such as this suggesting the existence of highly advantageous kinematic conditions.

A limitation of the methodology of the model development is that it cannot generate values for the localized forces experienced on the actual wing. The physiology of the guillemot wing ultimately evolved for flight and as such is a relatively lightweight and delicate structure initially optimized for aerial locomotion. When retracted the feathers do overlap and hence reinforce the overall strength of the wing, but overall remains relatively fragile.

Clearly the retracted wing shape helps in reducing the overall power requirement when considering the combination of inertial and hydrodynamic contributions as demonstrated by the results presented in table 2, but at this stage of the results the authors are unable to provide values for wing loading. It is likely that the retracted wing shape opted for by the guillemot is motivated by the need to reduce this wing loading experienced to a value that they can take. It can be concluded however that the inertial contributions experienced by the wing are less when retracted over the range of kinematics when generating 1 N of thrust. Although not presented within this paper, this trend has been found to be true for other values of thrust and forward velocity.

## 4. Conclusions

The goal of this work was to lay a mathematical framework to understand the manner in which the common guillemot achieves this duality of locomotion in both aerial and aquatic substrates, with the broad vision of providing insights into potential design solutions for future robotic vehicles.

The biological example has several key variations in the flapping technique adopted in air and water. Aside from the variations in flapping kinematics, the key mechanism utilized by the common guillemot, and indeed other avian species capable of aerial and aquatic locomotion, is the retraction of the wing reducing the overall wing plan form area and moment arm about which the wing is flapped.

Empirically determined values associated with the profile drag of the common guillemot confirm a potential 50% reduction in profile drag due to the retraction of the wing. Using these determined values along with previously determined values associated with parasitic drag [10], the overall drag associated with the guillemot is found to be lower in water than values calculated by theory for aerial locomotion.

Using an evolutionary algorithm within a mission-based optimization process, simulated behaviour based on the common guillemot has provided initial validation of the model, where by suggested wing geometry based on a mission reproducing the typical habits of the guillemot leads to the suggestion of wing geometry and flapping kinematics close in comparisons to the actual biological example.

Through the use of the numerical model, the retracted wing formation is found to consistently reduce the power requirement of the flapping motion compared to the extended wing, highlighting the benefit of the folding arrangement. Inertial dynamic contributions are also found to be consistently lower for the retracted wing.

Future work aims to conduct more experimental work in order to validate the numerical model. The investigation of a retractable wing design and light-weight driving mechanisms for the flapping motion are also to be investigated, in light of the need for an overall light-weight vehicle structure.

## Acknowledgment

The authors thank the Bristol Robotics Laboratory (BRL) for the use of their rapid prototyping machine. Thanks also go to Ed Charles from the Natural History Unit, BBC Bristol for providing footage and images that have been a great help during the research. This research was supported by the UK Engineering and Physical Sciences Research Council (EPSRC) and its Doctoral Training Assistantship (DTA) Program.

## References

[1] Ifju P G, Jenkins D A, Ettinger S, Lian Y, Shyy W and Waszak M R 2002 Flexible-wing-based micro air vehicles *40th AIAA Aerosp. Sci. Meet. Exhibit (Reno)* AIAA 2002-0705 pp 1–13

[2] Kingsley D A 2005 A cockroach inspired robot with artificial muscles *PhD Thesis* Case Western Reserve University, Cleveland, OH

[3] Licht S 2008 Biomimetic oscillating foil propulsion to enhance underwater vehicle agility and maneuverability *PhD Thesis* MIT/WHOI, Cambridge, USA

[4] Schroer R T, Boggess M J, Bachmann R J, Quinn R D and Ritzmann R 2004 Comparing cockroach and whegs robot body motions *IEEE Int. Conf. on Robotics and Automation (New Orleans, LA, USA)*

[5] Boria F J, Bachmann R J, Ifju P G, Quinn R D, Vaidyanathan R, Perry C and Wagner J 2005 A sensor platform capable of aerial and terrestrial locomotion *IEEE Int. Conf. on Intelligent Robots and Systems (Edmonton, Canada)* pp 3288–93

[6] Crespi A, Badertscher A, Guignard A and Ijspeert A J 2005 *AmphiBot I*: an amphibious snake-like robot *Robot. Auton. Syst.* **50** 163–75

[7] Ribak G and Weihs D 2004 How do cormorants counter buoyancy during submerged swimming? *J. Exp. Biol.* **207** 2101–14

[8] Heath J P *et al* 2006 Regulation of stroke pattern and swim speed across a range of current velocities: diving by common eiders wintering in polynyas in the Canadian Arctic *J. Exp. Biol.* **209** 3974–83

[9] Johansson L C and Norberg U M L 2001 Lift-based paddling in diving grebe *J. Exp. Biol.* **204** 1687–96

[10] Hui C A 1988 Penguin swimming. I. Hydrodynamics *Physiol. Zool* **61** 333–43

[11] Johansson L C and Aldrin B S W 2002 Kinematics of diving Atlantic puffins (*Fratercula arctica* L.): evidence for an active upstroke *J. Exp. Biol.* **205** 371–8

[12] Lovvorn J R, Croll D A and Liggins G A 1999 Mechanical versus physiological determinants of swimming speeds in diving Brunnich's guillemots *J. Exp. Biol.* **202** 1741–52

[13] Gaston A J 1998 *The Auks: Bird Families of the World* (Oxford: Oxford University Press)) *Series* 5

[14] Pennycuick C J 1987 Flight of auks (Alcidae) and other northern seabirds compared with southern procellariiformes: ornithodolite observations *J. Exp. Biol.* **128** 335–47

[15] Norberg U M 1990 Vertebrate flight: mechanics, physiology, morphology, ecology and evolution *Zoophysiology* vol 27 (Berlin: Springer)
[16] Lovvorn J R, Liggins G A, Borstad M H, Calisal S M and Mikkelsen J 2001 Hydrodynamic drag of diving birds: effects of body size, body shape and feathers at steady speeds *J. Exp. Biol.* **204** 1547–57
[17] Lock R J, Vaidyanathan R and Burgess S C 2008 Morphing modes of mobility in natural and engineered systems *Proc. Design and Nature IV (Algarve, Portugal, June)* pp 157–67
[18] accessed 15 May 2008 http;//www2.ups.edu/biology/museum/COMUwing1.jpg
[19] Stettenheim P 1959 Adaptations for underwater swimming in the common murre (*Uria aalge*) *PhD Thesis* University of Michigan, Ann Arbor, MI
[20] Ellington C P 1984 The aerodynamics of hovering insect flight: II. Morphological parameters *Phil. Trans. R. Soc.* B **305** 17–40
[21] Fossen T I 1999 *Guidance and Control of Ocean Vehicles. Chichester* (West Sussex: Wiley)
[22] Read D A, Hover F S and Triantafyllou M S 2003 Forces on oscillating foils for propulsion and manoeuvring *J. Fluids Struct.* **17** 163–83
[23] Techet A H 2008 Propulsive performance of biologically inspired flapping foils at high Reynolds numbers *J. Exp. Biol.* **211** 274–9
[24] Georgiades C, Nahon M and Buehler M 2009 Simulation of an underwater hexapod robot *Ocean Eng.* **36** 39–47
[25] Abbott H and Von Doenhoff A E 1959 *Theory of Wing Sections* (New York: Dover)
[26] Lock R J, Vaidyanathan R and Burgess S C 2010 Development of a biologically inspired multi-modal wing model for aerial-aquatic robotic vehicles *IEEE Int. Conf. on Intelligent Robots and Systems (Taipei, Taiwan)*
[27] Lock R J, Vaidyanathan R and Burgess S C 2010 Mission based optimization of a biologically inspired multi-modal wing model for aerial-aquatic robotic vehicles *11th Conf.—Towards Autonomous Robotic Systems (Plymouth, UK)* pp 140–7